→ Langchain

Langchain — Framework / library

↳ Develop Applications powered by LLM's

↓

Bridge the gap b/w raw LLM's model App



Langchain — Features → hen AI

→ Prompt templates
→ Document loaders
→ Parsers
→ memory
→ Agents
→ Third party integration → DB, vector DB, Tool, External API
etc. ...

LLM's → plug & play

Lang chain
↓
Language

↳ Pipeline
series of Modules
series of steps

step1 → step2 → step3

connect the series of steps

0.

LangChain Ecosystem

```
LangChain Core
├── langchain-openai (provider integration
├── langchain-anthropic
└── langchain-community
LangGraph (the future)
├── Agent orchestration
├── State management
└── Production deployment
LangSmith (observability)
├── Debugging & monitoring
├── Evaluation & testing
└── Performance optimization
```

← Agentic system

← Monitoring / Debugging

Template ←

Dear ——

Respected ——

pipeline operator → |

↳ prompt | llm

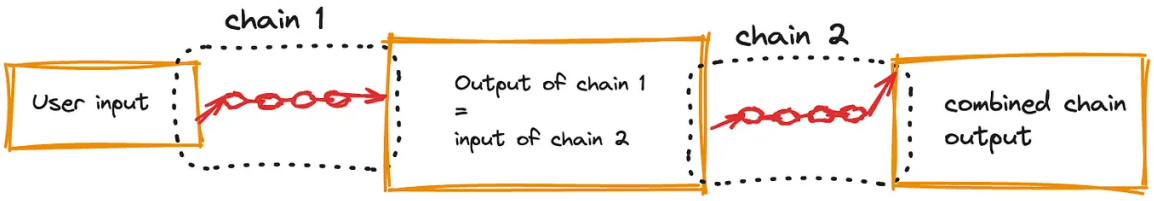
LCEL → Langchain Expression Languge

customer service BOT

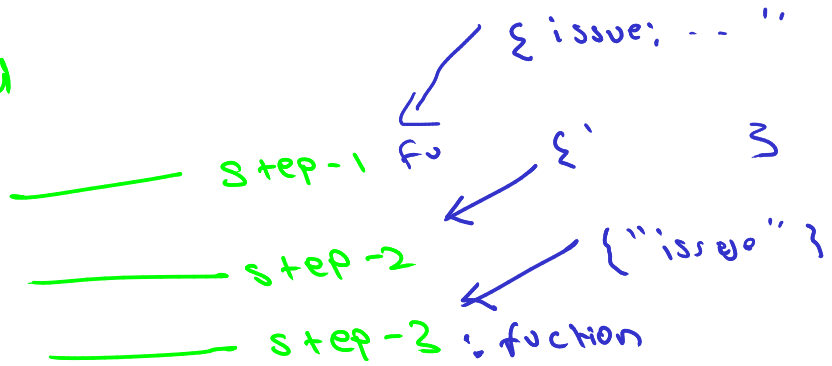
T1 Sentiment

T2 Classify (Type of Issue)

T3 Suggesting as solution / write professional Email



Runnable Parallel



Runnable lambda ✓

Runnable branch ✓

Invoice ←

product ✓

quantity ✓

price ✓

Runnable branch → Condional based Routing to specific chain

stream

batch